2. Technology Stack

Area	What is used	Where	Purpose
Frontend framework	React 18.3.1 with TypeScript	src/App.tsx, src/main.tsx, src/components/*	Builds the interactive dashboard and expense workflow.
Build tool	Vite 6 with @vitejs/plugin-react	vite.config.ts, package.json	Runs the dev server and creates static production assets in dist/.
Styling	Tailwind CSS 3, PostCSS, custom CSS	tailwind.config.js, postcss.config.js, src/index.css	Responsive UI, theme switching, chart animations, dark-mode polish.
Icons	lucide-react	Buttons, cards, scan panel, chart controls	Adds consistent UI icons such as Sparkles, Camera, Trash, Pencil, Calendar.
AI logic	Local rule-based categorization and insight engine	src/utils/aiEngine.ts	Suggests categories, confidence, reasons, practical actions, and spending insights.
OCR	Tesseract.js 6	src/components/ReceiptScanPanel.tsx	Reads text from receipt photos/screenshots or live camera captures.
Storage	Browser localStorage plus React state	src/App.tsx, src/lib/auth.ts	Stores demo users/current user/theme; expenses are initialized from localStorage and managed in app state.
Desktop app	macOS .app bundle + LaunchAgent + Python/Node HTTP server	ExpenseAI.app/Contents/*	Lets the project open like a normal Mac app without manually running terminal commands.

3. Runtime Architecture

Backend status

Current project backend: no traditional backend API. The app is a static React build. In development, Vite serves it. In the desktop app, a local HTTP server serves the built files from ExpenseAI.app/Contents/Resources/site.

1. User opens ExpenseAI.app or the Vite dev URL.
2. The browser loads index.html and bundled React assets.
3. React mounts App.tsx, checks localStorage for currentUser, theme, and existing expense data.
4. If no user is active, Login.tsx shows the login/sign-up form.
5. After login, the dashboard renders stats, expense form, expense list, insights, and charts.
6. Receipt images are processed in the browser by Tesseract.js; parsed results are copied into the expense form.
7. The app computes AI suggestions, insights, category totals, and chart slices locally from the expense array.